HANDS-ON

SmartCliff
Career Mobility Solutions

Hands-On No. : 1

Topics Covered : Introduction to C Programming

Date : 29-06-26

Solve the following problems

Q. No.	Question Detail	Level
1	Neelam wants to share her code with a colleague, who may modify it. Thus, she wants to include the date of the program creation, the author and other with the program. What component should she use? a) header files b) Iteration **c) Comments** d) Preprocessor **Explanation:** Comments are used to write notes for humans, like author name, creation date, and purpose of the program. They are ignored by the compiler, so they don't affect the code. This makes comments the correct place to store such information.	Basic
2	A 10-bit unsigned integer has the following range: a) 0 to 1000 b) 0 to 1024 **c) 0 to 1023** d) 0 to 1025 **Explanation:** An unsigned binary number has the range formula: 0 to (2^n - 1), where *n* is the number of bits.	Basic

	For 10 bits: (2^ {10} = 1024), so the range becomes 0 to 1024 − 1 = 1023. That's why the maximum value is 1023.	
3	By default a real number is treated as a a) float **b) double** c) long double d) Depends on the memory model you are using. **Explanation:** In C, any real number like 3.14 or 9.81 is automatically treated as a double unless you add a suffix like f. This is because double provides higher precision, and the C standard prefers accuracy by default.	Basic
4	As boolean can store only two values(i.e, 1 and 0), so tricha can store a list of binary data in boolean datatype. A data type is stored as a 6 bit signed integer. Which of the following cannot be represented by this data type? a) -12 b) 6 c) 18 **d) 32** **Explanation:** A signed number with 6 bits has the range formula: −2^(n−1) to +2^(n−1) − 1. So its range is: $-(2^5)$ to $(2^5 - 1)$ → −32 to +31. So values like −12, 6, and 18 fit inside this range, but 32 is too big, so it cannot be stored.	Basic
5	A language has 28 different letters in total. Each word in the language is composed of maximum 7 letters. You want to create a data-type to store a word of this language. You decide to store the word as an array of letters. How many bytes will you assign to	Basic

	the data-type to be able to store all kinds of words of the language. **a) 7** b) 75 c) 29 d) 126 **Explanation:** There are 28 letters, so one letter needs 1 byte to store (because 1 byte can store up to 256 values). A word can have maximum 7 letters, so you need 7 bytes to store the longest possible word. Therefore, the datatype must be 7 bytes.	
6	Parul takes as input two numbers: a and b. a and b can take integer values between 0 and 255. She stores a, b and c as 1-byte data type. She writes the following code statement to process a and b and put the result in c. c = a + 2*b To her surprise her program gives the right output with some input values of a and b, while gives an erroneous answer for others. For which of the following inputs will it give a wrong answer? a) a = 200 b = 10 **b) a = 10 b = 200** c) a = 50 b = 100 d) a = 100 b = 50 **Explanation:** Each variable is 1 byte, so it can store only 0 to 255.The expression is: c = a + 2*b. For a = 10, b = 200 → 2*b = 400, and 10 + 400 = 410, which is greater than 255, so it overflows and gives a wrong result. The other options give results ≤ 255, so they work correctly.	Basic

7	What statement will you add to the following code snippet if it is to produce the output given below? 31.240000 6.360000 5.460000 float a = 31.24; double b = 6.36; long double c = 5.46; **Solution:** printf("%f %lf %Lf",a,b,c); **Explanation:** Each data type has its own format specifier in printf, otherwise garbage values or wrong output will appear. float must be printed with %f double must be printed with %lf long double must be printed with %Lf Using all three in the correct order prints the values exactly as: 31.240000 6.360000 5.460000.	Basic
8	What is short int in C programming? a) Basic datatype of C b) Qualifier c) **short is the qualifier and int is the basic datatype** d) All of the mentioned	Basic

9	Which of the following is the correct output for the program given below? void main(){ float a = 0.7; if(a<0.7) printf("C"); else printf("C++"); } **a) C** b) C++ c) Compiler reports an error saying a float cannot be compared with a double. d) None of the above	Basic
10	Which of the following is the correct output for the program given below? void main(){ float a = 0.7; if(a<0.7f) printf("C"); else printf("C++"); } a)C **b)C++** c) Compiler reports an error saying a float cannot be compared	Basic

	with a double. d) None of the above	
11	We want to round off x, a float, to an int value. The correct way to do so will be **a) y = (int)(x + 0.5);** b) y = int (x + 0.5); c) y = (int)x + 0.5; d) y = (int)((int)x + 0.5)	Basic
12	Which of the following is the correct output for the program given below? #include<stdio.h> #include<math.h> void main(){ float n = 1.54; printf("%f %f", ceil(n),floor(n)); } **a) 2.000000 1.000000** b) 1.500000 1.500000 c) 1.550000 2.000000 d) 1.000000 2.000000	Basic
13	Which of the following is the correct output for the program given below? #include<stdio.h> void main(){ printf("%d %d %d " ,sizeof(3.14f) ,sizeof(3.14) ,sizeof(3.14l)); } a) 4 4 4 b) 4 8 8 **c) 4 8 10** d) 4 8 12	Basic
14	To print out a and b given below, which of the following printf() statement will you use?	Basic

	float a = 3.14; double b = 3.14; **a) printf("%f %lf",a,b);** b) printf("%Lf %f",a,b); c) printf("%Lf %Lf",a,b); d) printf("%f %Lf",a,b);	
15	**What is the output of this program?** #include <stdio.h> int main() { int i = 0, j = 1, k = 2, m; m = i++ || j++ || k++; printf("%d %d %d %d", m, i, j, k); return 0; } a) 1 1 2 3 **b) 1 1 2 2** c) 0 1 2 2 d) 1 2 3 3	
16	In the context of the following printf() in C, pick the best statement. i) printf("%d",8); ii) printf("%d",090); iii) printf("%d",00200); iv) printf("%d",0007000); a) Only i) would compile. And it will print 8. b) Both i) and ii) would compile. i) will print 8 while ii) will print 90 c) All i), ii), iii) and iv) would compile successfully and they will print 8, 90, 200 & 7000 respectively. d) **Only i), iii) and iv) would compile successfully. They will print 8, 128 and 3584 respectively.**	Basic

<table>
<tr><td>17</td><td>Suspose a, b, c and d are int variables. For ternary operator in C (? :), pick the best statement.
a) a>b ? : ; is valid statement i.e. 2nd and 3rd operands can be empty and they are implicitly replaced with non-zero value at run-time.
b) a>b ? c=10 : d=10; is valid statement. Based on the value of a and b, either c or d gets assigned the value of 10.
c) a>b ? (c=10,d=20) : (c=20,d=10); is valid statement. Based on the value of a and b, either c=10,d=20 gets executed or c=20,d=10 gets executed.
d) All of the above are valid statements for ternary operator.</td><td>Basic</td></tr>
<tr><td>18</td><td>What will be the output of the following?
#include <stdio.h>
int main(void)
{
int a;
int i = 1;
int b = 10 * i + sizeof(--i) + 4 - 10 / i;
printf("%d", b);
return 0;
}
a) 4
b) 2
c) 6
d) 8</td><td>Basic</td></tr>
<tr><td>19</td><td>What will be the output of the following program?
#include <stdio.h>
int main(void)
{
int i = 40 >> 5 << 3 >> 2 << 1;
printf("%d", i);
return 0;
}</td><td>Basic</td></tr>
</table>

	a) **4** b) 0 c) 40 d) 1	
20	**What will be the output of the following program?** #include <stdio.h> int main(void) { int i = 10 > 9 > 7 < 8; printf("%d", i); return 0; } a) **1** b) 20 c) 10 d) 0	Basic
21	**What will be the output of the following program?** #include <stdio.h> int main(void) { int x = 4, y = 4, z = 4; (x == y == z) ?printf("Hello"): printf("World"); return 0; } 1. Hello 2. 0 3. 1 **4. World**	Basic
22	**What will be the output of the following program?** #include <stdio.h> int main(void) { int x = 10, y = 15;	Basic

	x ^= y ^= x ^= y; printf("%d%d", x, y); return 0; } 1. 44 **2. 1510** 3. 55 4. 45	
23	**What will be the output of the following program?** #include <stdio.h> int main(void) { int a; int i = 4; a = 24 || --i; printf("%d %d", a, i); return 0; } **1. 1 4** 2. 4 4 3. 4 1 4. 1 1	Basic
24	**What will be the output of the following program?** #include <stdio.h>> int main() { int y = 10; (y++ > 9 && y++ != 10 && y++ > 11) ? printf("%d", y) : printf("%d", y); return 0; } a) 11 b) 12	Basic

	c) 13 d) none of above	
25	**What will be the output of the following program?** #include <iostream> using namespace std; int main() { int a = 3, b = 5, c, d; c = a, b; d = (a, b); printf("c=%d d=%d", c, d); return 0; } **a) c=3 d=5** b) c=5 d=5 c) can't be determined d) none of these	Basic
26	**What is the output of the following program?** #include <stdio.h> int main() { unsigned int i = 0x80; printf("%d ", i << 1); return 0; } a) 0 **b) 256** c) 100 d) 80	
27	**What is the output of the following program?** #include <stdio.h> int main() {	

	unsigned int a = 0xffff; unsigned int k = ~a; printf("%d %d\n", a, k); return 0; } a) **65535 -65536** b) -65535 65535 c) 65535 65535 d) -65535 -65535	
28	**What is the output of the following program?** #include <stdio.h> int main() { unsigned int a = 0xffff; unsigned int k = ~a; printf("%d %u\n", a, k); return 0; } a) 65535 -65536 b) -65535 65535 c) **65535 4294901760** d) -65535 -65535	
29	**What Is The Output Of this program?** #include <stdio.h> int main() { unsigned int a = -1; unsigned int b = 4; printf("%d\n", a << b); return 0; } a) -1 b) 16	

	c) 8 **d) -16**	
30	**What Is the Output Of this program?** #include <stdio.h> int main() { unsigned int m = 32; printf("%d %d", m >> 1, ~m); return 0; } a) 16 5 b) 64 -32 c) 16 33 **d) 16 -33**	
31	**What Is the Output Of this program?** #include <stdio.h> int main() { int x; x = 5 > 8 ? 10 : 1 != 2 < 5 ? 20 : 30; printf("%d", x); return 0; } a) 20 b) **30** c) 1 d) 0	
32	**What is the output of the program?** #include <stdio.h> int main() { int x; x = 5 < 8 ? 1 != 2 < 5 == 0 ? 10 : 20 : 30;	

<table>
<tr><td></td><td>printf(":%d", x);
return 0;
}
a) 10
b) 20
c) 30
d) 1</td><td></td></tr>
<tr><td>33</td><td>What is the output of the program?
#include <stdio.h>
#include <stdio.h>
int main()
{
int x;
x = 2 > 5 != 1 ? 5 < 8 && 8 > 2 ? !5 ? 10 : 20 :
30 : 40;
printf("Value of x:%d", x);
return 0;
}
a)10
b)20
c)30
d)40</td><td></td></tr>
<tr><td>34</td><td>What is the output of following program?
#include <stdio.h>
int main()
{
int a = 1;
int b = 1;
int c = a || --b;
int d = a-- && --b;
printf("a = %d, b = %d, c = %d, d = %d", a, b, c, d);
return 0;
}</td><td></td></tr>
</table>

	a) a = 0, b = 1, c = 1, d = 0 b) **a = 0, b = 0, c = 1, d = 0** c) a = 1, b = 1, c = 1, d = 1 d) a = 0, b = 0, c = 0, d = 0	
35	**What is the output of following program?** #include <stdio.h> int main() { int x = 10; int y = (x++, x++, x++); printf("%d %d\n", x, y); return 0; } a) **13 12** b) 13 13 c) 10 10 d) compiler Dependent	
36	**What is the output of this C code?** #include <stdio.h> int main() { int var; /*Suppose address of var is 2000 */ void *ptr = &var; *ptr = 5; printf("var=%d and *ptr=%d",var,*ptr); return 0; } a) It will print "var=5 and *ptr=2000" b) It will print "var=5 and *ptr=5" c) It will print "var=5 and *ptr=XYZ" where XYZ is some random address **d) Compile error**	Basic
37	**What does the following C code do?** int main() { int arr[] = {1, 2, 3};	Basic

	int *p = arr; printf("%d", *(p + 1)); return 0; } a. Prints the first element of arr **b. Prints the second element of arr** c. Prints the third element of arr d. Error	
38	**Find the error:** int main() { int arr[3] = {5, 10, 15}; int *p = arr; printf("%d", *p + 2); return 0; } a. Pointer arithmetic error b. Syntax error **c. No error** d. Wrong printf format	Basic
39	What will be the output of the following C code? #include <stdio.h> void main() { char *s = "hello"; char *p = s; printf("%c\t%c", *(p + 1), s[1]); } a. h e b. e l c. h h **d. e e**	Basic

40	**What will be the output of the following C code?** #include <stdio.h> void main() { char *a[10] = {"hi", "hello", "how"}; int i = 0, j = 0; a[0] = "hey"; for (i = 0;i < 10; i++) printf("%s", a[i]); } a. hihellohow Segmentation fault **b. hihellohow followed by 7 null values** c. heyhellohow followed by 7 null values d. heyhellohow Segmentation fault	Basic
41	**Spot the mistake:** struct Data { int val; }; int main() { struct Data *ptr; ptr->val = 100; printf("%d", ptr->val); return 0; } a. Uninitialized pointer used b. Syntax error c. No error d. Wrong printf format **Answer: No error**	Basic
42	**What will be the output of the following C code?** #include <stdio.h> struct point { int x; int y;	Basic

	}; void foo(struct point*); int main() { struct point p1[] = {1, 2, 3, 4}; foo(p1); } void foo(struct point p[]) { printf("%d\n", p[1].x); } a. Compile time error b. 3 c. 2 d. 1 **Answer: B** **Explanation :** The array p1 is initialized with {1, 2, 3, 4}, where each struct point consists of two integers. Thus, p[1].x refers to the x value of the second struct point, which is 3.	
43	**What will be the output of the following C code on a 64-bit system?** #include <stdio.h> struct student { char *c; }; void main() { struct student s[2]; printf("%d", sizeof(s)); }	Basic

	a. 2 b. 4 c. 16 d. 8 **Answer: c** **Explanation:** On a 64-bit system, size of pointer is 8 bytes. Here, we are printing the size of an array of 2 structures, hence, the size will be 2×8 = 16 bytes.	
44	**What will be the output of the following C code?** #include <stdio.h> struct student { char *c; }; void main() { struct student m; struct student *s = &m; s->c = "hello"; printf("%s", s->c); } a. Hello b. Run time error c. Nothing d. Depends on compiler **Answer: A** **Explanation:** The pointer s->c is assigned the string "hello", and printf("%s", s->c); correctly prints the string "hello".	Basic
45	**What will be the output of the following C code?** #include <stdio.h> struct p	Basic

	`` { int x[2]; }; struct q { int *x; }; int main() { struct p p1 = {1, 2}; struct q *ptr1; ptr1->x = (struct q*)&p1.x; printf("%d\n", ptr1->x[1]); } ``` a. Compile time error b. Segmentation fault/code crash c. 2 d. 1 **Answer: C**	
46	**What will be the output of the following C code?** ``` #include <stdio.h> struct point { int x; int y; }; void foo(struct point*); int main() { struct point p1[] = {1, 2, 3, 4, 5}; foo(p1); ```	Basic

	`` } void foo(struct point p[]) { printf("%d %d\n", p->x, (p + 2)->y); } ``` a. Compile time error b. 1 0 c. 1 somegarbagevalue d. undefined behaviour **Answer: B** **Explanation**: The array p1 is initialized with values {1, 2, 3, 4, 5}, but it only initializes two struct point objects. The first struct point is {1, 2}, and the second is {3, 4}. When foo(p1) is called, p->x is 1. The expression (p + 2)->y accesses the y value of the third struct point, which does not exist and results in 0 due to how the array is accessed in memory. Thus, the output is 1 0	
47	**What will be the output of the following C code?** ``` #include <stdio.h> int main() { struct p { char *name; struct p *next; }; struct p *ptrary[10]; struct p p, q; p.name = "xyz"; p.next = NULL; ptrary[0] = &p; q.name = (char*)malloc(sizeof(char)*3); ```	Basic

<table>
<tr><td></td><td>strcpy(q.name, p.name);
q.next = &q;
ptrary[1] = &q;
printf("%s\n", ptrary[1]->next->next->name);
}

a. Compile time error
b. Depends on the compiler\
c. Undefined behaviour
d. xyz

Answer: D
Explanation: The struct p instance q is initialized with name set to a copy of p.name ("xyz") and next pointing to itself. This creates a self-referential loop for q. As a result, ptrary[1]->next->next->name resolves to q.name, which is "xyz", so the program prints "xyz".</td><td></td></tr>
<tr><td>48</td><td>What will be the output of the following C code?
#include <stdio.h>
struct p
{
unsigned int x : 1;
unsigned int y : 1;
};
int main()
{
struct p p;
p.x = 1;
p.y = 2;
printf("%d\n", p.y);
}

a. 1</td><td>Basic</td></tr>
</table>

	b. 2 c. 0 d. Depends on the compiler **Answer: C** **Explanation**: In the given code, struct p defines two bit-fields x and y, each of size 1 bit. When p.y = 2; is assigned, the value 2 is truncated to fit within the 1-bit size limit of y. This truncation results in p.y holding the value 0. Therefore, the program prints 0 when printf("%d\n", p.y); is executed.	
49	**What will be the output of the following code?** struct example { int a : 2; int b : 2; }; int main() { struct example e = {3, 2}; printf("%d %d\n", e.a, e.b); return 0; } a) 3 2 b) -1 -2 c) 1 0 d) 0 1 **Answer:** b) -1 -2 (because 3 and 2 cannot fit in 2 bits, so they wrap around and get stored as negative values due to sign extension)	Basic
50	**What will be the output of the following code?** union data { int i; float f; char str[20]; };	Basic

	int main() { union data d; d.i = 10; printf("%d\n", d.i); d.f = 220.5; printf("%d\n", d.i); return 0; } a) 10 and 0 b) 10 and 220 c) 10 and some undefined value d) Compilation error **Answer:** c) 10 and some undefined value **Explanation:** After assigning d.f = 220.5, it overwrites the memory previously occupied by d.i because both share the same memory location. Accessing d.i after setting d.f results in undefined behavior.	
51	**What is the size of the following union on a system where int is 4 bytes and double is 8 bytes?** union mix { int a; double b; char c[9]; }; a) 9 bytes b) 12 bytes c) 16 bytes d) 18 bytes **Answer:** c) 16 bytes **Explanation:** The size of a union is determined by the size of its largest member. double is 8 bytes, and char c[9] would require 9 bytes, but to align it properly, the total size becomes 16 bytes.	Basic

52	**What will be the size of the following union assuming int is 4 bytes and char is 1 byte?** union data { struct { char a; int b; } s; int x; }; a) 4 bytes b) 5 bytes c) 8 bytes d) 16 bytes **Answer:** c) 8 bytes **Explanation:** The size of the union will be the size of the largest member. The structure contains a char (1 byte) and an int (4 bytes), but padding is added to align int, resulting in 8 bytes. The int x is 4 bytes but the structure is larger.	Basic